A Systematic Review of Intelligent Multilingual and Multimodal AI Systems for Cross-Disability Assistive Accessibility

Abhinav Papini[1*], Sai Ritesh Domakuntla[1], Arbaz Ahmed[1], Manikanta Pendela[1], Anjusha Dr.[1]

[1*]Department of Computer Science and Engineering, VNR Vignana Jyothi Institute of Engineering and Technology (VNR VJIET), Hyderabad, Telangana, India.

*Corresponding author(s). E-mail(s): papiniabhinav@gmail.com;
Contributing authors: sairiteshdomakuntla@gmail.com;
arbazahmed1729@gmail.com; pendelamanikanta6205@gmail.com;
anjusha_p@vnrvjiet.ac.in;

Abstract

The use of assistive technologies among people with sensory and speech disabilities is hampered by the fragmentation of architecture, single-modality interfaces, and linguistic bias in non-English speaking populations. The existing commercial assistive systems can only perform isolated actions such as optical character recognition (OCR), text-to-speech, and single-turn image captioning and cannot accommodate the users with compound disabilities. This paper reviews the development of intelligent assistive technologies in the domains of visual assistance, multilingual speech interaction, live video stream, and privacy-preserving AI. We review the seminal papers in the domain of vision assistance, multilingual large language models (MLLMs), multilingual speech recognition for Indic languages, and edge-cloud video streaming systems. Some critical research gaps are discussed, which include the absence of a unified cross-disability system, exclusion of the population of Indic language speakers from multilingual models, latency bottleneck in live scene narration, and lack of privacy design in the continuous assistive feed. Based on the identified gaps, this work introduces a unified assistive system validated through experimental benchmarking results: live narration latency of 688.3 ms over Wi-Fi, OCR word error rate of 4.2% for printed signages, and voice command classification accuracy above 94% in English, Hindi, and Telugu.

Keywords: Multimodal AI, Cross-Disability Accessibility, Vision-Language Models, Multilingual Speech Recognition, Real-Time Scene Narration, Privacy-Preserving AI

1 Introduction

More than 1.3 billion people around the world—about 16% of the world's population—are affected by disabilities [1]. Of these people, individuals affected by sensory and speech problems constitute a particularly vulnerable group who need rapid, precise, and context-sensitive assistive technologies to function indepen dently. People suffering from visual impairments face problems in reading sig nages, medicine warnings, and navigation hazards. People suffering from deafness or hearing loss require instantaneous visual or haptic representations of sound events. People with speech or motor impairments are often locked out of voice-based digi tal systems that do not have alternative communication methods [2, 3].

Contemporary assistive tools suffer from four documented structural shortcomings [4]: (i) **Modal Fragmentation**—users must switch between disparate applications for OCR, object localization, captioning, and conversation; (ii) **Linguistic Bias**—tools favour English while excluding hundreds of millions of Indic language speakers [5]; (iii) **Latency Bottlenecks**—cloud-based inference introduces multi-second delays unsuitable for real-time navigation; and (iv) **Privacy Vulnerabilities**— continuous camera feeds are archived by cloud platforms without explicit user consent.

1.1 Objectives

The objectives of this research work, each mapped to a defining keyword, are:

1. [**Cross-Disability Unification**] To design and evaluate a unified assistive framework that simultaneously addresses visual (*blind/low-vision*), auditory (*deaf/hard-of-hearing*), vocal (*speech-impaired*), and cognitive accessibility needs within a single interaction model.
2. [**Multilingual Indic Support**] To implement an on-device, zero-latency phonetic command engine covering English, Hindi (Devanagari and Romanized), and Telugu (native script and Romanized), addressing the linguistic exclusion documented across Indian-language ASR research [11].
3. [**Real-Time Scene Narration**] To achieve continuous live scene narration with end-to-end latency below 850 ms through a WebSocket streaming pipeline, single-concurrency frame queuing, and multi-model fallback cascade [12].
4. [**Multimodal Vision Intelligence**] To embed on-demand high-resolution vision. Multimodal large language models (LLMs) can be used for visual inspection and live narration. (MLLMs) [9] that can perform optical character recognition (OCR), object localization, and contextual scene understanding.

5. [**Privacy-by-Design**] To design a 'zero-persistence' ephemeral processing pipeline where no camera frame, transcript, or biometric tokens are stored in any version of the system [3].
6. [**Multi-Turn Conversational Grounding**]To facilitate multi-turn visual dialogue with state information such that the user can pose additional questions on an already-seen image.

The remainder of this paper is organized as follows. Section 2 presents the literature survey. Section 3 identifies research gaps. Section 4 describes the methodology. Section 5 presents comparative experiments. Section 6 concludes the paper, followed by future scope in Section 7.

2 Literature Survey

Assistive computing has evolved through multiple paradigms, from rule-based image filters to deep convolutional networks and contemporary multimodal transformer architectures. Below we survey the four principal research pillars.

2.1 Vision-Assistive Applications and Scene Understanding

The VizWiz platform was first used by Bigham et al. [2] for crowd-sourced visual assistance, allowing blind users to take pictures and receive answers almost immediately from sighted volunteers. The research conveyed a crucial point: pictures taken by blind users show a greater degree of blurring, poor framing and poor lighting than the non-blind users, and over 30% of queries had no standard benchmark images that automated systems could answer because of these characteristics. These findings proved that there was a need for models that provide uncertainty estimates, not confidence values.

Microsoft's Seeing AI [6] enhanced the modularity that was part of the CNN architecture by introducing individual sub-modes (short text, barcode, scene, face) for iOS. However, each mode is an independent inference pipeline which does not have semantic continuity, because the system cannot answer a query about an object that was detected in the previous frame.

Google Lookout [7] provided on-device continuous object and text detection via TensorFlow Lite, with snapshot latencies of 0.8–1.5 s. Envision AI [13] expanded wearable visual recognition to smart glasses with snapshot latencies of 2–4 s; however, multi-turn dialogue and Indic language support are missing in both systems.

Guo et al. [17] conducted a systematic benchmark of assistive mobile vision systems, finding that real-world error rates were 2–4× higher than controlled laboratory benchmarks due to environmental variability. Howard et al. [20] have proved that lightweight mobile architectures like MobileNet are able to drastically reduce inference latency on mobile hardware, while Jocher et al. [21] have demonstrated that object detection is possible in real-time (above 100 frames per second) on edge devices using YOLO-family detectors. But both architectures do not provide for rich, context-rich natural language scene description.

2.2 Multimodal Large Language Models

The Transformer architecture [14] paved the way for all current multimodal models. Brown et al. [18] showed that few-shot learning is powerful in language models at scale across diverse tasks. Later, OpenAI [19] introduced GPT-4V, the first commercially released MLLM that enables vision and text reasoning in a production pipeline.

Gemini Team [9] released the Gemini family of natively multimodal models, which can process interleaved text, images, and video in a single token space. They found that their method outperforms task-specific CNN pipelines by a wide margin for zero-shot scene description, dense OCR, and spatial reasoning, and inference is performed with under 500 ms latency on stable internet connections, even for the lightweight Flash-Lite variant.

Liu et al. [10] introduced LLaVA, an open-weights instruction-tuned vision-language model that is fine-tuned from LLaMA using multimodal instruction data generated using GPT-4. In their ablation studies, visual instruction tuning achieved significantly better performance than zero-shot prompting on scene description tasks, providing a standardized reproducible open-source baseline for accessibility research.

Be My Eyes [8] adopted GPT-4V as a "Virtual Volunteer", suggesting that MLLMs can replace human sighted volunteers for simple visual queries. However, their deployment remains English-only, closed-source, and reliant on synchronous REST calls introducing 3.5–6.0 s latency—unsuitable for real-time navigation.

2.3 Multilingual Speech Recognition and Indic Language Processing

Radford et al. [15] presented Whisper, a multilingual ASR model trained via large-scale weak supervision on 680,000 hours of audio data across 96 languages. Their results showed that Whisper achieves near-human Word Error Rate (WER) on many languages but exhibits substantially elevated WER on low-resource Indic languages due to limited training representation.

Jawahar et al. [5] explored multilingual transformer performance on Indian languages, finding that standard transformer architectures trained predominantly on English data suffer 30–60% degradation in token-level accuracy when applied to Hindi, Telugu, Tamil, and Bengali. Their analysis attributed this to morphological complexity, script diversity, and insufficient pre-training corpora.

Kumar et al. [11] evaluated speech recognition for low-resource Indian languages, finding that WER scores were 2–5× higher than English baselines, even for high-resource languages such as Hindi. They noted that code-mixed speech such as Hinglish and Tenglish poses particularly severe challenges due to within-utterance script switching.

Ramesh et al. [22] released the Sangraha dataset and trained IndicConformer and IndicWhisper models specifically for Indian languages, achieving substantial WER reductions. Tang et al. [16] demonstrated that multilingual pre-training with extensible vocabulary expansion enables zero-shot cross-lingual transfer, which is directly applicable to low-resource Indic command parsing.

2.4 Real-Time Streaming and Edge-Cloud Architectures

The WebSocket protocol (RFC 6455) was established by Fette and Melnikov [12], setting the groundwork for persistent full-duplex browser-server communication. Their protocol analysis found that WebSockets decrease per-message overhead from 871 bytes (HTTP polling) to 2 bytes (WebSocket frame header), allowing high-frequency frame transmission with low bandwidth cost.

Morris [4] offered an extensive overview of AI accessibility opportunities and challenges, stating that latency, cost, and model hallucination are the three main limitations preventing practical deployment. He also highlighted the danger of MLLMs producing confident but incorrect spatial descriptions—a critical safety issue for blind navigation users.

The W3C [3] released WCAG 2.2 with 87 accessibility success criteria for web content. Generally, WCAG 2.2 covers UI-level accessibility (e.g., screen reader support, color contrast, and keyboard navigation) but offers no guidance on AI-generated content quality, ephemeral data handling, or streaming inference architectures—a normative gap this research directly addresses.

3 Research Gap

The literature reviewed in Section 2 reveals the following unresolved gaps that collectively motivate this research work:

1. **No unified cross-disability platform exists.** None of the reviewed commercial products—Seeing AI [6], Lookout [7], Envision AI [13], Be My Eyes [8]—goes beyond visual impairments. Users with multiple disabilities (e.g., simultaneous vision and speech deficits) have no single solution that handles them. No academic prototype reviewed integrates visual, auditory, vocal, and cognitive assistance within a single coherent system.
2. **Indic languages are absent from all major assistive AI products.** Although there are 500 M+ Hindi and 80 M+ Telugu speakers, no assistive platform offers on-device phonetic command parsing in native Indic scripts or Romanized transliteration. As noted by Kumar et al. [11] and Jawahar et al. [5], off-the-shelf ASR exhibits 2–5× higher WER on Indian language speech, so purely cloud-based ASR is unreliable for safety-critical voice commands.
3. **No system guarantees live narration with a validated sub-second latency.** Current solutions have latencies of 1.8–6.0 s per frame using REST calls, significantly above the practical threshold for pedestrian navigation. To the best of our knowledge, no published work provides an empirically validated end-to-end narration pipeline with freshest-frame queueing, single-concurrency enforcement, and a multi-model cascade achieving below 1 s across real mobile network conditions.
4. **In the absence of privacy architecture, assistive streaming feeds remain exposed.** Cloud-based assistive platforms archive continuous camera feeds and transcripts. Neither WCAG 2.2 [3] nor any reviewed system specifies or validates an ephemeral, zero-persistence processing model for assistive AI video streams.

5. **Multi-turn visual dialogue is unsupported.** All scene-understanding systems (Seeing AI, Lookout, Envision AI) operate in single-turn snapshot mode. No system retains conversational context to ground follow-up queries to prior visual responses, limiting natural interaction for users who ask sequential questions about a scene.

Table 1 summarises these gaps against all reviewed systems.

Table 1 Capability comparison of existing systems against this research work's contributions. ✓ = Supported, ∼ = Partial, × = Not supported.

System	Cross-Disab.	Indic Lang.	<1 s Live	Zero-Persist.	Multi-Turn
Seeing AI [6]	×	×	×	×	×
Lookout [7]	×	×	∼	×	×
Envision AI [13]	×	×	×	×	×
Be My Eyes [8]	×	×	×	×	✓
Voice Assistants	×	∼	∼	×	∼
This Research Work	✓	✓	✓	✓	✓

4 Proposed Methodology

The proposed three-tier client-edge-cloud assistive framework consists of a **Client Tier** (React Native mobile application), an **Orchestration Gateway** (Node.js Master Agent), and a **Multimodal AI Cloud** (MLLM inference backends). Each architectural tier is designed to directly resolve one or more research gaps identified in Section 3.

4.1 Cross-Disability Adaptive Profiling

Onboarding involves the selection of a need vector $N \in \{\text{visual, hearing, speech, general}\}$. This profile is saved locally and sent with each inference request. The Orchestration Gateway dynamically compiles the AI system prompt via:

$$\mathcal{S}(L,\, N) \;=\; \mathcal{P}_{\text{base}} \;\oplus\; \mathcal{T}_{\text{lang}}(L) \;\oplus\; \bigoplus_{n \,\in\, N} \mathcal{H}_{\text{need}}(n) \tag{1}$$

where $\mathcal{P}_{\text{base}}$ is the universal base prompt, $\mathcal{T}_{\text{lang}}(L)$ injects the user's language instruction ($L \in \{\text{en-US, hi-IN, te-IN}\}$), and $\mathcal{H}_{\text{need}}(n)$ applies need-specific constraints: egocentric spatial coordinates for visual impairment, suppression of auditory references for hearing impairment, and verbosity reduction for speech impairment.

4.2 WebSocket Live Scene Narration Pipeline

To achieve Objective 3 (real-time narration), the system implements a freshest-frame WebSocket pipeline formalised in Algorithm 1.

Algorithm 1 Freshest-Frame Single-Concurrency Streaming Algorithm

Require: Camera frame F_t received at time t; model cascade $\mathcal{M}$; concurrency flag `busy`; freshest-frame buffer $\mathcal{B}$
Ensure: Narration R_t emitted within latency budget $T_{\max} = 850\,\text{ms}$

if `busy` $= \top$ **then**
  $\mathcal{B} \leftarrow F_t$ ▷ Replace outdated frame; no queue growth
else
  `busy` $\leftarrow \top$
  $F \leftarrow \mathcal{B}$ if $\mathcal{B} \neq \emptyset$ else F_t
  for each model $m \in \mathcal{M}$ **do**
    $R \leftarrow \textsc{Infer}(m, F)$
    if R is valid **then**
      emit R to client; **break**
    end if
  end for
  `busy` $\leftarrow \bot$
end if

The three-tier model cascade is ordered by latency:

$$\mathcal{M} = \langle \underbrace{\texttt{gemini-2.5-flash-lite}}_{\mu\approx 486\,\text{ms}}, \underbrace{\texttt{gemini-2.5-flash}}_{\mu\approx 821\,\text{ms}}, \underbrace{\texttt{gemini-flash-latest}}_{\text{fallback}} \rangle \quad (2)$$

The total end-to-end latency budget is:

$$T_{\text{total}} = T_{\text{cap}} + T_{\text{net,up}} + T_{\text{infer}} + T_{\text{net,dn}} + T_{\text{TTS}} \quad (3)$$

4.3 Multilingual Phonetic Command Engine

Voice commands are categorized using an on-device normalized phonetic pattern matcher. Input transcripts are normalized to Unicode NFC format and stripped of punctuation (including the Devanagari *danda* '—'), and then matched against a multi-script token tree. The command categorization function is:

$$\mathcal{C}(u) \;=\; \arg\max_{a \in \mathcal{A}} \sum_{p \in \mathcal{P}_a} \mathbb{1}\big[\text{norm}(u) \supseteq p\big] \quad (4)$$

where $\mathcal{A}$ is the action set, $\mathcal{P}_a$ is the set of phonetic patterns for action a, and $\mathbb{1}[\cdot]$ is the indicator function. Unmatched utterances default to `followup`, forwarding the natural-language query directly to the conversational MLLM pipeline.

The action vocabulary covers both native scripts and Romanized transliterations:

$$\mathcal{A} = \{\texttt{describe}, \texttt{read}, \texttt{capture}, \texttt{find}(x), \texttt{stop}, \texttt{repeat}, \texttt{followup}(t)\} \quad (5)$$

4.4 Intent Routing and Multi-Turn Dialogue

The Master Agent dispatches incoming messages through a deterministic intent parser:

$$\mathcal{I}(m) = \begin{cases} \text{translation} & \text{if } m \text{ matches translation lexemes} \\ \text{vision-query} & \text{if context image } \exists \text{ and } m \in Q \\ \text{help} & \text{if } m \text{ requests operational guidance} \\ \text{chat} & \text{otherwise (follow-up turn)} \end{cases} \tag{6}$$

Active capability registry $\mathcal{C} = \{\text{vision} : \top,\ \text{speech} : \top,\ \text{translation} : \bot\}$ gates dispatch; inactive capabilities return graceful degradation messages.

4.5 Privacy-Preserving Ephemeral Processing

Frame data is encoded into in-memory base64 buffers on-device, transferred via TLS-encrypted WebSockets, processed only in volatile server-side RAM, and marked for garbage collection immediately after inference. Formally, for any frame F_t:

$$\forall t :\ \Pr\big[\textsc{Persist}(F_t)\big] = 0 \tag{7}$$

Image buffers, transcripts, and biometric tokens are never persisted to any storage at any system tier, achieving a stronger guarantee than the data minimisation guidelines of WCAG 2.2 [3].

5 Experimental Results and Comparative Analysis

This chapter will include empirical analysis of the system in four different dimensions: end-to-end latency, OCR accuracy, multilingual command accuracy, and live translation. Comparison of results with previous systems will be provided where available.

5.1 Experimental Setup

All benchmarks were conducted on a physical Android device (Snapdragon 8 Gen 2, 12 GB RAM) connected to a Node.js backend server (8-core Intel Xeon, 16 GB RAM). Network conditions simulated were: (i) stable Wi-Fi at 50 Mbps, and (ii) mobile 4G LTE at 15 Mbps with 60 ms round-trip jitter. Inference backends used the Gemini 2.5 Flash-Lite model as primary and Gemini 2.5 Flash as fallback.

5.2 Latency Analysis

The end-to-end latency budget decomposes as:

$$T_{\text{total}} = T_{\text{cap}} + T_{\text{net,up}} + T_{\text{infer}} + T_{\text{net,dn}} + T_{\text{TTS}} \tag{8}$$

Table 2 details mean and 95th-percentile latencies across 100 consecutive requests per condition.

Table 2 End-to-end latency breakdown (ms) across pipeline stages and network conditions.

Stage	Wi-Fi Mean	Wi-Fi P95	4G Mean	4G P95
Frame Capture & Encode	48.2	62.0	51.4	68.5
WebSocket Ingestion	4.1	6.8	8.3	14.2
Flash-Lite Inference	485.6	620.4	512.0	688.0
Flash Inference (Fallback)	820.5	1140.0	864.2	1210.5
Network RTT	38.4	55.2	142.6	210.4
On-Device TTS Init	112.0	145.0	118.0	152.0
Live Narration (Flash-Lite)	**688.3**	**889.4**	**832.3**	**1133.1**
High-Res REST Inspection	**1420.0**	**1890.2**	**1745.5**	**2280.0**

Comparison. Existing systems report REST-based snapshot latencies: Seeing AI 1.8–3.2 s, Be My Eyes 3.5–6.0 s, Envision AI 2.0–4.0 s, and Google Lookout 0.8–1.5 s (on-device only, no MLLM). This research work's WebSocket Flash-Lite pipeline at 688.3 ms under Wi-Fi represents a **2.6**× latency reduction versus the best prior REST-based MLLM deployment (Be My Eyes, $\mu \approx 3.5$ s), and achieves sub-second latency even under 4G LTE—a condition untested and unreported by any prior system.

5.3 OCR and Text Extraction Accuracy

Text extraction was evaluated on 120 standardised scenes spanning street signage, printed medication labels, handwritten notes, and restaurant menus under two illumination levels: standard (300 lux) and low-light (30 lux).

Performance is measured by Word Error Rate (WER) and Character Error Rate (CER):

$$\text{WER} = \frac{S + D + I}{N_w}, \qquad \text{CER} = \frac{S_c + D_c + I_c}{N_c} \tag{9}$$

where S, D, I denote substitutions, deletions, and insertions at the word level, S_c, D_c, I_c at the character level, and N_w, N_c are reference lengths.

Table 3 OCR accuracy comparison: WER (%) across systems and scene types. Lower is better. (–) = not reported by the system.

System	Printed Signs	Medication Labels	Handwritten	Low-Light
Seeing AI [6]	8.4	12.1	22.3	28.6
Google Lookout [7]	6.2	9.8	–	19.4
GPT-4V (Be My Eyes) [19]	5.1	7.4	14.2	17.8
This Research Work	**4.2**	**6.8**	**12.6**	**11.4**

Under adequate illumination, this research work achieves a WER of 4.2% on printed signage and 6.8% on curved medication containers. Under low-light (30 lux), the system maintains WER of 11.4%—a **36**% reduction over the next-best reported system (GPT-4V, 17.8%). This improvement is attributed to the adaptive system prompt

that instructs the model to prefix uncertainty disclosures ("*The image is dim, but I can make out...*") rather than hallucinating confident but erroneous text.

5.4 Multilingual Voice Command Accuracy

The on-device phonetic command engine was evaluated across 300 voice command trials (100 per language) spoken by native speakers with regional accents, including code-mixed utterances.

Table 4 Voice command classification accuracy (%) by language and command class.

Command Class	English	Hindi (Dev/Translit)	Telugu (Native/Translit)	Macro Avg
`describe`	99.0	96.0	95.0	96.7
`read`	98.0	97.0	94.0	96.3
`find`	97.0	94.0	92.0	94.3
`stop`	100.0	99.0	98.0	99.0
`repeat`	99.0	96.0	95.0	96.7
`followup`	96.0	93.0	91.0	93.3
Overall	**98.2**	**95.8**	**94.2**	**96.1**

Most importantly, the safety-critical `stop` command achieves >98% accuracy irrespective of the language used. This on-device engine provides zero network latency for command classification—a direct advantage over any cloud-ASR pipeline. No prior assistive system reports accuracy for Indic language commands; thus, no direct comparison is possible, which itself underscores Gap 2 identified in Section 3.

5.5 Live Navigation and Translation (Objectives 2 and 3 Realised)

This research work accomplishes two objectives relevant to the survey:

Live Scene Navigation (Objective 3): The system continuously narrates the user's environment through the WebSocket pipeline, offering navigational cues ("A staircase begins two steps ahead on your left", "Glass door ahead, handle on the right") based on the spatial reasoning performed by the MLLM. In tests conducted in an office building, the system correctly identified 91% of static obstacles and 78% of dynamic obstacles (moving persons, swinging doors) within the 850 ms budget.

Multilingual Translation (Objective 2 partial): The intent router detects translation requests and passes text extracted via the OCR pipeline to the MLLM with an explicit translation instruction. In trials with 50 Hindi and 50 Telugu signage images, translation accuracy (BLEU-4) was 0.74 for Hindi and 0.71 for Telugu, outperforming baseline Google Translate API (BLEU-4: 0.68 / 0.62) on domain-specific signage text containing local proper nouns and abbreviations.

6 Conclusion

This review paper systematically surveyed the state of intelligent assistive technologies across four research domains: vision-assistive applications, multimodal large language models, multilingual speech recognition for Indic languages, and real-time edge-cloud streaming architectures. Through this analysis, five critical research gaps were identified and formalised: the absence of unified cross-disability platforms, linguistic exclusion of Indic language speakers, latency barriers in live scene narration, insufficient privacy architectures in assistive feeds, and the lack of multi-turn conversational grounding in visual assistance systems.

This research work presents a directly responsive framework that addresses each identified gap with empirically validated solutions. The freshest-frame WebSocket pipeline with single-concurrency enforcement and multi-model cascade achieves end-to-end live narration latencies of 688.3 ms (Wi-Fi) and 832.3 ms (4G LTE), a 2.6× improvement over the best prior MLLM-based deployment. On-device OCR achieves a 4.2% WER on printed signage and 11.4% under low-light conditions. The multilingual phonetic command engine achieves 98.2% (English), 95.8% (Hindi), and 94.2% (Telugu) command accuracy—the first reported Indic language command accuracy results in the assistive AI literature. The zero-persistence ephemeral architecture guarantees $\Pr[\textsc{Persist}(F_t)] = 0$ for all captured camera frames, offering a stronger privacy guarantee than any reviewed architecture.

Collectively, these results show that advanced multimodal AI can be architected to serve multilingual and cross-disability populations when inclusivity is treated as a primary architectural constraint rather than an afterthought. The empirical evidence presented establishes a reproducible baseline for future research in inclusive, real-time assistive AI.

7 Future Scope

Building on the findings and limitations of this research work, the following directions are identified for future investigation:

1. **On-Device SLM Integration.** Integrating quantized Small Language Models (e.g., Gemma 2B via MediaPipe) for fully offline scene description will eliminate the network dependency critical in rural or low-connectivity environments. Achieving sub-500 ms on-device inference quality comparable to cloud MLLMs remains an open challenge in model compression.
2. **Spatial Haptic Navigation.** Extending the output modalities to include wearable haptic arrays (vibration vests or gloves) will serve DeafBlind users and users in acoustically inappropriate environments. Designing learnable, non-fatiguing somatosensory encodings for dynamic obstacle geometry is a significant HCI research challenge.
3. **Extended Indic Language Coverage.** Expanding the phonetic command engine to Tamil, Kannada, Malayalam, Marathi, Bengali, and Odia will bring coverage to over 95% of India's linguistically diverse population. Each language requires

dedicated multi-script token trees for native script, Romanized transliteration, and code-mixed colloquial speech.

4. **Symbol Boards with the help of AI.** Combining predictive AAC symbol boards into the mobile application will be helpful for non-verbal users who rely on symbol selection rather than voice for communication. Predicting the conversational context can be very effective in reducing the mental effort involved in navigating symbol boards.
5. **Differential Privacy through Visual Anonymization.** Blurring faces and anonymizing license plates on-device before they are sent to the cloud will provide formal differential privacy guarantees for bystanders who may appear incidentally in the video stream during assistive navigation sessions.
6. **Longitudinal Field Studies.** Controlled laboratory assessments are to be supplemented by longitudinal deployment studies with real disability communities in naturalistic settings—offices, markets, public transport—to expose failure modes and usability barriers not obvious under laboratory conditions.

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- **Ethics Approval:** Not applicable. Voice command trials were conducted with informed consent from volunteer participants.
- **Data Availability:** Evaluation scripts and benchmark results are available in the project repository.
- **Code Availability:** https://github.com/sairiteshdomakuntla/Major-Project
- **Author Contributions:** Abhinav Papini led OCR pipeline design and benchmarking. Sai Ritesh Domakuntla designed the overall system architecture and streaming pipeline. Arbaz Ahmed developed the multilingual speech engine. Manikanta Pendela built the backend infrastructure. Dr. Anjusha supervised the research, methodology, and manuscript review.

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